



MT1003 – Calculus and Analytical Geometry

Assignment No: 01

Individual Assignment

Section: All BSCS Sections

Semester: Fall 2023

Due date: Thursday, 14 September 2023, 01:00pm

Marks: 16*10=160

Instructions:

1. Plagiarized work will result in zero marks.
 2. No retake or late submission will be accepted.
 3. Attach complete code, results, and screenshot for questions that require programming solution. Programs/codes should be typed.
 4. Questions with  icon require partial or complete solution using MATLAB.
 5. **The assignment is to be submitted in softcopy as well as in hardcopy. Submit the hardcopy before the deadline through CR, and softcopy on GCR.**
 6. The softcopy should be a single PDF file of your complete assignment including programming and non-programming questions.
 7. The PDF file should be according to the following format: id_section_A1 e.g., i23-123456_A_A1. A1 in the end denotes Assignment 1. **The first page must include complete student information, including name, section, id, course name, and assignment number.**
 8. The images of by-hand solution should be properly scanned. You can use any mobile application such as CamScanner or Adobe Scan for scanning. Each of these applications allow you to export pdf or image files which you can use to combine with your programming solutions. Do not attach direct images from the camera application of your mobile phone, or screenshots.
 9. MATLAB is the only approved programming tool. Octave-Online is also allowed.
-



Question 1

You are designing a rectangular garden next to a wall. The garden has two vertical sides, one of which is the wall, and a flat top. The flat top is made of concrete that costs \$8 per square foot, and the other two sides are made of wooden fencing, which costs \$4 per linear foot. You have a budget of \$800 for the garden's construction. Express the width 'w' as a function of the length 'l' of the garden. Also find the domain of the function.

Solution

You are designing a rectangular garden next to a wall. The garden has two vertical sides, one of which is the wall, and two horizontal sides. In this configuration, there are two horizontal sides and one vertical side (the wall), and the other vertical side is open.

Given:

- Cost of concrete top (flat top): \$8 per square foot.
- Cost of wooden fencing: \$4 per linear foot.
- Budget constraint: \$800.

Let 'w' represent the width of the rectangular garden in feet, and 'l' represent the length of the garden in feet.

1. The cost of the concrete top (flat top) is $8 * (w * l) = 8wl$ dollars.
2. The cost of the two horizontal wooden sides (fencing) is $4 * 2 * w = 8w$ dollars, where 2 represents the two horizontal sides.
3. One of the vertical sides is the wall, so there's no additional cost for fencing along that side.

Now, let's express the total cost C as a function of 'w' and 'l':

$$C(w, l) = 8wl + 8w$$

You have a budget of \$800, so we can set up an equation for the budget constraint:

$$C(w, l) \leq 800$$

Now, substitute the expression for C(w, l):

$$8wl + 8w \leq 800$$

Let's factor 'w' out of the terms:

$$w(8l + 8) \leq 800$$

Now, divide both sides by $(8l + 8)$:

$$w \leq 800 / (8l + 8)$$



Simplify the right side:

$$w \leq 100 / (l + 1)$$

So, the function ' $w(l)$ ' that represents the width ' w ' as a function of the length ' l ' is:

$$w(l) = 100 / (l + 1)$$

Now, let's find the domain of this function:

1. ' w ' should be non-negative because it represents the width, so:

$$w(l) \geq 0$$

2. The denominator ' $l + 1$ ' should not be equal to zero:

$$l + 1 \neq 0$$

Now, let's analyze the domain:

- ' w ' should be non-negative: $100 / (l + 1) \geq 0$

Since the denominator is always positive ($l + 1 > 0$), ' w ' is non-negative for all real values of ' l '.

- The denominator ' $l + 1$ ' should not be zero:

$$l + 1 \neq 0$$

$$l \neq -1$$

So, the domain of the function is all positive real numbers.



Question 2



Taxicabs such as Uber and Lyft have a unique fare structure. Passengers are charged a flat fee of \$3.00 for entering a cab and then an additional \$0.75 for each quarter of a mile (or fraction thereof) traveled. Ignoring any additional charges, if 'x' represents the distance traveled in miles, write a function $C(x)$ that represents the cost of the taxi fare.

Using MATLAB, sketch the graph of $C(x)$. Then, based on the graph of the fare function, find each of the following limits, if they exist:

- | | | | |
|------|---------------------------------------|---------------------------------------|-------------------------------------|
| i) | a) $\lim_{x \rightarrow 0.30^-} C(x)$ | b) $\lim_{x \rightarrow 0.30^+} C(x)$ | c) $\lim_{x \rightarrow 0.30} C(x)$ |
| ii) | a) $\lim_{x \rightarrow 0.45^-} C(x)$ | b) $\lim_{x \rightarrow 0.45^+} C(x)$ | c) $\lim_{x \rightarrow 0.45} C(x)$ |
| iii) | a) $\lim_{x \rightarrow 0.75^-} C(x)$ | b) $\lim_{x \rightarrow 0.75^+} C(x)$ | c) $\lim_{x \rightarrow 0.75} C(x)$ |

Solution

1. **Base Fare:** \$3.00

2. **Distance-Based Charges:** \$0.75 for each quarter of a mile (or fraction thereof) traveled. So, the distance-based charge is $0.75 \cdot (4x) = 3x$.

Now, let's create the function:

$$C(x) = \text{Base Fare} + \text{Distance-Based Charges}$$

$$C(x) = 3.00 + 3x$$

This function represents the total cost of the taxi fare for a given distance traveled (x) in miles.

Now, let's calculate the limits based on the graph of $C(x)$ using MATLAB:

To sketch the graph of $C(x)$ in MATLAB, you can use the following code:



matlab

Copy code

```
x = 0:0.01:2; % Range of x values
C = 3 + 3*x;   % Cost function

% Plot the cost function
plot(x, C)
xlabel('Distance Traveled (miles)')
ylabel('Cost of Taxi Fare ($)')
title('Taxi Fare vs. Distance Traveled')
grid on
```

After plotting the graph, you can visually examine the behavior of the function near the specified values of x to find the limits.

Now, let's calculate the limits:

i) For $x = 0.30$ miles:

a) $\lim_{x \rightarrow 0.30^-} C(x)$: This is the limit as x approaches 0.30 from the left.

$$\lim_{x \rightarrow 0.30^-} C(x) = \lim_{x \rightarrow 0.30^-} (3 + 3x) = 3 + 3(0.30) = 3.90$$

b) $\lim_{x \rightarrow 0.30^+} C(x)$: This is the limit as x approaches 0.30 from the right.

$$\lim_{x \rightarrow 0.30^+} C(x) = \lim_{x \rightarrow 0.30^+} (3 + 3x) = 3 + 3(0.30) = 3.90$$

c) $\lim_{x \rightarrow 0.30} C(x)$: This is the overall limit as x approaches 0.30.

$$\lim_{x \rightarrow 0.30} C(x) = 3.90$$



iii) For $x = 0.75$ miles:

a) $\lim_{x \rightarrow 0.75^-} C(x)$: This is the limit as x approaches 0.75 from the left.

$$\lim_{x \rightarrow 0.75^-} C(x) = \lim_{x \rightarrow 0.75^-} (3 + 3x) = 3 + 3(0.75) = 5.25$$

b) $\lim_{x \rightarrow 0.75^+} C(x)$: This is the limit as x approaches 0.75 from the right.

ii) For $x = 0.45$ miles:

a) $\lim_{x \rightarrow 0.45^-} C(x)$: This is the limit as x approaches 0.45 from the left.

$$\lim_{x \rightarrow 0.45^-} C(x) = \lim_{x \rightarrow 0.45^-} (3 + 3x) = 3 + 3(0.45) = 3.135$$

b) $\lim_{x \rightarrow 0.45^+} C(x)$: This is the limit as x approaches 0.45 from the right.

$$\lim_{x \rightarrow 0.45^+} C(x) = \lim_{x \rightarrow 0.45^+} (3 + 3x) = 3 + 3(0.45) = 3.135$$

c) $\lim_{x \rightarrow 0.45} C(x)$: This is the overall limit as x approaches 0.45.

$$\lim_{x \rightarrow 0.45} C(x) = 3.135$$



Question 3

You are an automotive engineer developing a fuel-efficient hybrid car, the "EcoDrive 2024." The car's MPG (miles per gallon) varies with speed according to the following piecewise function:

$$\begin{cases} 100 - s & \text{if } s \leq 55\text{mph} \\ \frac{2200}{s} & \text{if } s > 55\text{mph} \end{cases}$$

1. Find the speed for maximum MPG in electric mode and with the gasoline engine.
2. Calculate the MPG at specific speeds like 45 mph, 60 mph, and 70 mph.
3. Determine the speed at which the car transitions from electric to gasoline mode.
4. Discuss whether the MPG function is continuous at the transition speed (55 mph) and when approaching it from different directions. Provide reasons for your conclusions.

Solution

1. Maximum MPG

In electric mode, the maximum MPG=100 occurs when $s=0$. However, this means that the car is not moving, and is not a practical situation.

In gasoline mode, the maximum MPG occurs at the transition speed, as MPG decreases with an increase in speed.

2. Calculate the MPG at specific speeds:

- At 45 mph ($s = 45$ mph), in electric mode:
 $M(45) = 100 - 45 = 55$ MPG
- At 60 mph ($s = 60$ mph), in gasoline mode:
 $M(60) = \frac{2200}{60} = 36.67$ MPG
- At 70 mph ($s = 70$ mph), in gasoline mode:
 $M(70) = \frac{2200}{70} = 31.43$ MPG

3. Determine the speed at which the car transitions:

The car transitions from electric to gasoline mode at $s = 55$ mph.



4. Analyze the continuity of the MPG function:

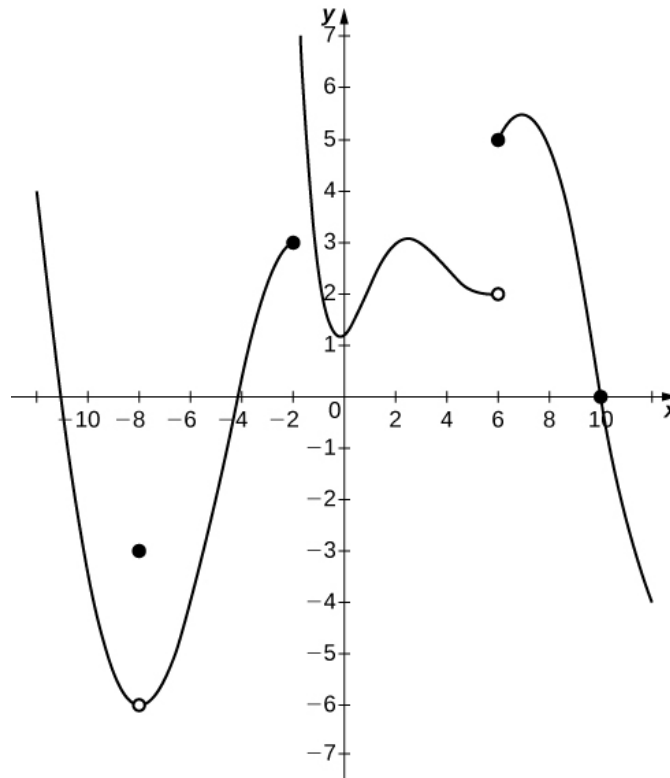
- The MPG function $M(s)$ is not continuous at $s = 55$ mph because the transition between electric and gasoline modes results in a discontinuity.
- When approaching $s = 55$ mph from the left side (electric mode), the MPG decreases linearly, approaching a limit of 0 MPG.
- When approaching $s = 55$ mph from the right side (gasoline mode), the MPG increases as speed decreases, approaching a maximum value of approximately 46.83 MPG.

So, the MPG function $M(s)$ has a jump discontinuity at $s = 55$ mph due to the mode transition.

Question 4

Use the graph of the function f to decide whether the value of the given quantity exists. If it does, find it. If not, explain why.

- $\lim_{x \rightarrow 10} f(x) = 0$
- $\lim_{x \rightarrow -2^+} f(x)$: vertical asymptote
- $\lim_{x \rightarrow -8} f(x) = -6$
- $\lim_{x \rightarrow 6} f(x)$: limit does not exist
- $f(-8) = -3$
- the vertical asymptotes of the graph of f : -2



Question 5

Classify the locations and types of discontinuities of the following functions.

- $f(x) = \frac{x^2}{x^2 - 1}$
- $f(x) = \text{greatest integer function}$
- $\frac{\sin(\pi x)}{x}$
- $f(x) = \begin{cases} x^2 - 2 & \text{if } x \neq -3 \\ 5 & \text{if } x = -3 \end{cases}$
- $f(x) = \begin{cases} e^x & x < 0 \\ x^2 & x \geq 0 \end{cases}$

Solution

- a) Discontinuity with an Asymptote at $x = -1$ and $x = 1$
- b) Jump discontinuity (non-removable discontinuity as the limits are finite and not-equal)
- c) Removable discontinuity
- d) There is removable discontinuity
- e) There is jump discontinuity

Question 6

Evaluate $\lim_{x \rightarrow 1} \arccos\left(\frac{1-\sqrt{x}}{1-x}\right)$.

Solution

$$\begin{aligned}\lim_{x \rightarrow 1} \arccos\left(\frac{1-\sqrt{x}}{1-x}\right) &= \arccos\left(\lim_{x \rightarrow 1} \frac{1-\sqrt{x}}{1-x}\right) \\&= \arccos\left(\lim_{x \rightarrow 1} \frac{1-\sqrt{x}}{(1-\sqrt{x})(1+\sqrt{x})}\right) \\&= \arccos\left(\lim_{x \rightarrow 1} \frac{1}{1+\sqrt{x}}\right) \\&= \arccos\left(\frac{1}{2}\right) \\&= \frac{\pi}{3}\end{aligned}$$



Question 7

Use the Squeeze Theorem to find the limit of $\lim_{x \rightarrow -\infty} \frac{5x^2 - \sin(3x)}{x^2 + 10}$.

Solution

$$-1 \leq \sin(3x) \leq +1 ,$$

$$-1 \leq -\sin(3x) \leq +1 ,$$

$$5x^2 - 1 \leq 5x^2 - \sin(3x) \leq 5x^2 + 1 ,$$

$$\frac{5x^2 - 1}{x^2 + 10} \leq \frac{5x^2 - \sin(3x)}{x^2 + 10} \leq \frac{5x^2 + 1}{x^2 + 10} .$$

$$\lim_{x \rightarrow -\infty} \frac{5x^2 - 1}{x^2 + 10}$$

$$= \lim_{x \rightarrow -\infty} \frac{5x^2 - 1}{x^2 + 10} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}}$$

$$= \lim_{x \rightarrow -\infty} \frac{5 - \frac{1}{x^2}}{1 + \frac{10}{x^2}}$$

$$= \frac{5 - 0}{1 + 0}$$

$$= 5 .$$



Question 8

A new pizzeria has opened near your home, the owner has decided to offer a 20% discount on the opening day on the pizza whose original price is x dollars. The owner is offering additional 15% discount coupon to those who agree to fill a feedback form afterwards. You and your friends have agreed to provide feedback and thus have become entitled to the additional 15% discount coupon. Use the concept of composite functions to identify the amount that you will pay after applying the coupon on the sale price.

Solution

Since the sale price is 20% off the original price, if an item is x dollars, its sale price is given by $f(x) = 0.80x$. Since the coupon entitles an individual to 15% off the price of any item, if an item is y dollars, the price, after applying the coupon, is given by $g(y) = 0.85y$. Therefore, if the price is originally x dollars, its sale price will be $f(x) = 0.80x$ and then its final price after the coupon will be $g(f(x)) = 0.85(0.80x) = 0.68x$.



Question 9

Scientists have developed a radioisotope-based power supply whose power output in watts is given by the following function:

$$P = 75e^{-\frac{t}{125}}$$

where t is the number of days the power supply is used. The experimental power supply has been installed in an unmanned undersea vehicle on an exploration mission. The vehicle requires 7watts to operate at full capacity. Use the given information to find the number of days for which the vehicle can operate at full capacity.

Solution

The solution for t is as follows:

$$7/75 = e^{-t/125}$$

$$\ln(7/75) = \ln(e^{-t/125})$$

$$\ln(7/75) = -t/125$$

$$t = -125 \ln(7/75) \approx 296.4$$

Question 10



The function $H(t) = 8\sin\left(\frac{\pi}{6}t\right)$ models the height H (in feet) of the tide t hours after midnight. Assume that $t = 0$ is midnight.

- Find the amplitude and period.
- Graph the function over one period by hand.
- Graph the function using MATLAB.
- What is the height of the tide at 4:30 a.m.?

Solution

Since the mathematical model of the height H (in feet) of the tide t hours after midnight given by

$$H(t) = 8\sin\left(\frac{\pi}{6}t\right) \quad (1)$$

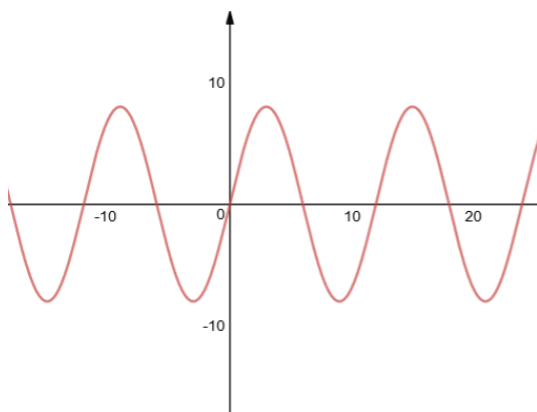
Compare with

$$f(x) = C + A\sin(B(x - \alpha))$$

where C is vertical shift, A is amplitude of f , B is changes the period

From (1), we get Amplitude = 8, and period given by

$$\begin{aligned} \text{Period} &= \frac{2\pi}{B} \\ &= 2\pi \frac{6}{\pi} \\ &= 12 \text{ hours} \end{aligned}$$



(c) The height of the tide at 4 : 30 a.m.

$$\begin{aligned} H(4.5) &= 8\sin\left(\frac{\pi}{6}(4.5)\right) \\ &= 5.57 \end{aligned}$$



Question 11



The local mass transit authority has given you the following passenger data to consider:

Year	1999	2000	2001	2002	2003	2004	2005	2006
Passengers (in millions)	76.4	78.6	81.5	87.8	88.4	92.4	94.0	95.0

Use MATLAB to plot the data and observe its trend. Find a linear function for the data and use it to predict the number of passengers for 2014.

Solution

We can choose any two of the data points to determine an equation. Let's use (2, 81.5) and (6, 94.0).

We first determine the slope of the line:

$$m = \frac{94.0 - 81.5}{6 - 2} = \frac{12.5}{4} = 3.125.$$

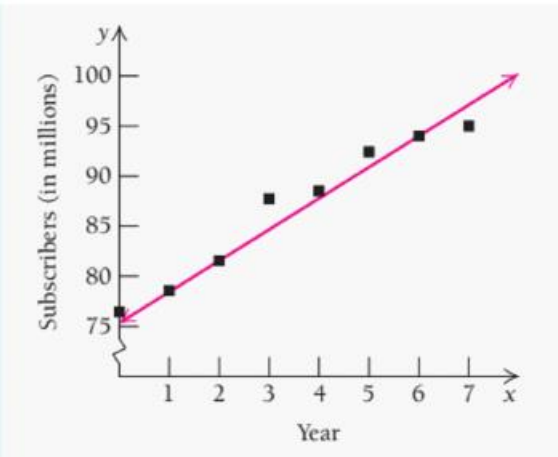
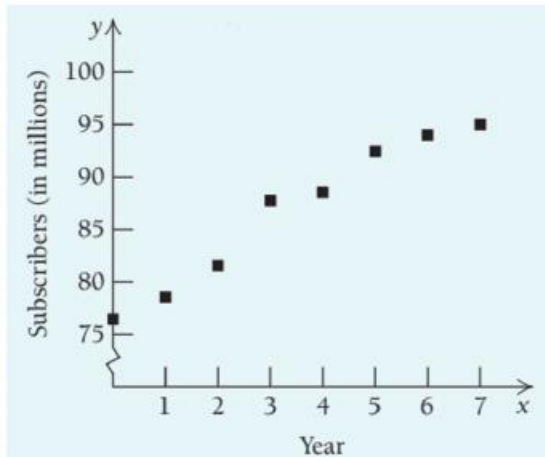
Then we substitute 3.125 for m and either of the points (2, 81.5) or (6, 94.0) for (x_1, y_1) in the point-slope equation. Using (6, 94.0), we get

$$y - 94.0 = 3.125(x - 6),$$

which simplifies to

$$y = 3.125x + 75.25,$$

where x is the number of years after 1999 and y is in millions. We can then graph the linear equation $y = 3.125x + 75.25$ on the scatterplot to see how it fits the data.



$$y = 3.125x + 75.25$$

Model

$$= 3.125(15) + 75.25$$

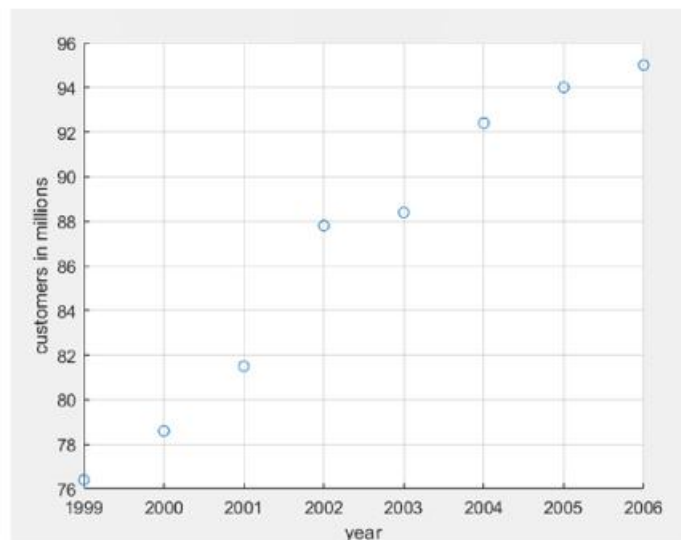
Substituting

$$\approx 122.1.$$

Rounding to the nearest tenth

We can predict that there will be about 122.1 million customers in 2014.

```
year=1999:2006;
customer=[76.4 78.6
81.5 87.8 88.4 92.4
94.0 95.0];
scatter(year, customer
)
grid on
ylabel('customers in
millions')
xlabel('year')
```



Question 12



Sketch the graphs of the functions, not by plotting points but by starting from basic function and applying transformations. Also plot the graphs using MATLAB and verify your answers.

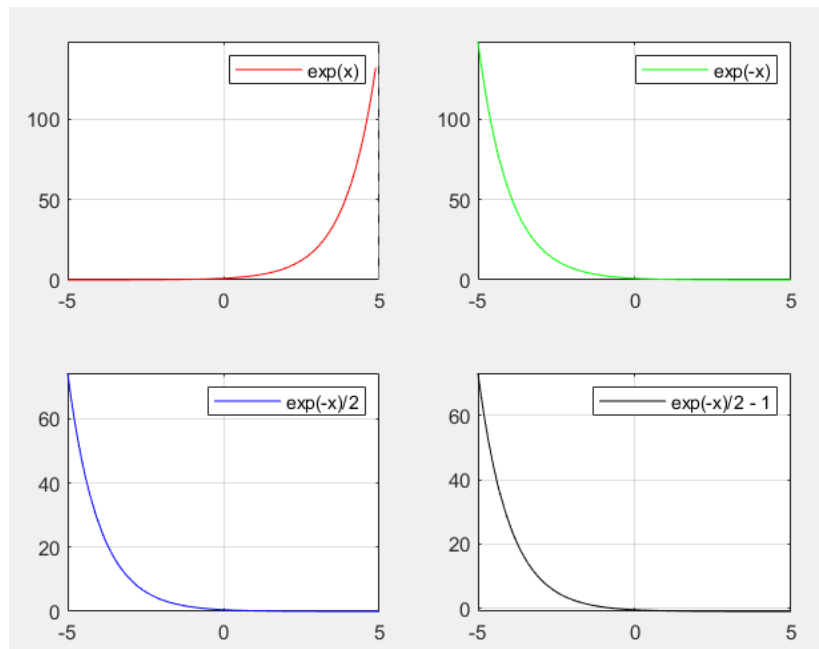
- a) $y = \frac{1}{2}e^{-x} - 1$
- b) $y = \ln(x - 2) - 1$
- c) $y = 2 + \sqrt{1 - (x - 1)^2}$

Solution

(12a) Draw parent function $y = e^x$. Reflect the graph about y-axis, then vertically compress the graph 0.5 units, and at the end shift the graph 1 unit down.

% Question 12a

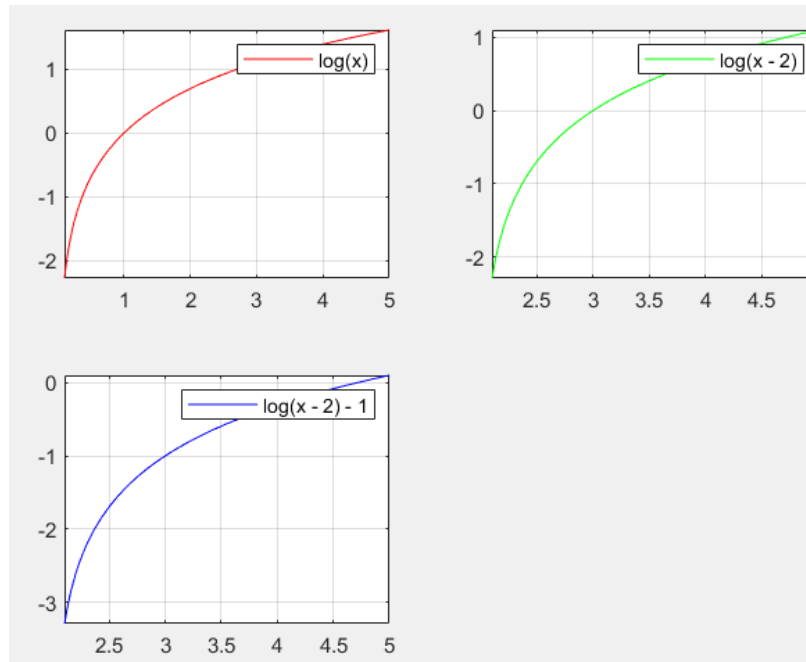
```
syms x
f=exp(x)
g=exp(-x)
h=(1/2)*exp(-x)
fn=(1/2)*(exp(-x))-1
subplot(2,2,1)
fplot(f,'r')
legend
grid on
subplot(2,2,2)
fplot(g,'g')
legend
grid on
subplot(2,2,3)
fplot(h,'b')
legend
grid on
subplot(2,2,4)
fplot(fn,'k')
legend
grid on
```



(12b) Draw parent graph $y = \ln(x)$. Then shift it 2 units right, and 1 unit down.

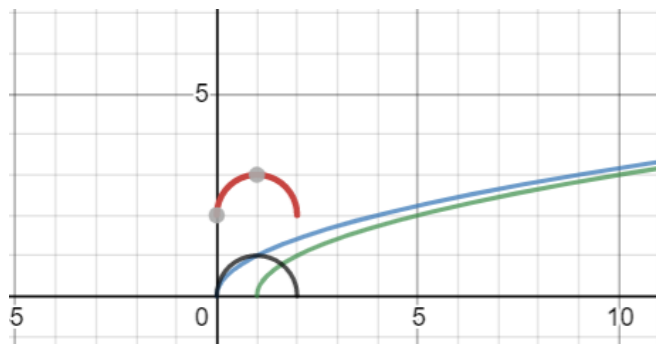
% Question 13b

```
syms x
f=log(x);
g=log(x-2);
h=log(x-2)-1;
subplot(2,2,1)
fplot(f,'r')
legend
grid on
subplot(2,2,2)
fplot(g,'g')
legend
grid on
subplot(2,2,3)
fplot(h,'b')
legend
grid on
```



(12c)

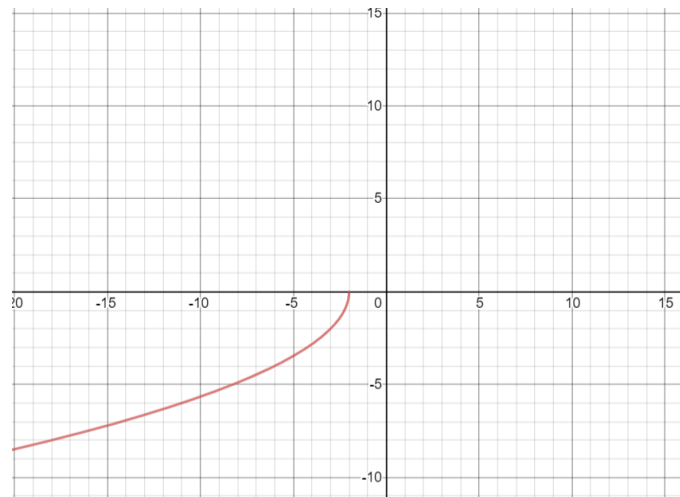
	$y = \sqrt{x}$
	$y = \sqrt{x-1}$
	$y = \sqrt{1-(x-1)^2}$
	$y = 2 + \sqrt{1-(x-1)^2}$



Question 13



Identify the base graph and all the transformations required to obtain the given graph from base graph. Verify the identified transformations using MATLAB. In the end you should obtain the function that will result in the given graph.



Solution

Base graph: $\sqrt{x}$

Transformations

1. $y = \sqrt{x - 2}$
2. $y = \sqrt{-x - 2}$
3. $y = -\sqrt{-x - 2}$
4. $y = -2\sqrt{-x - 2}$

Question 14

Identify the domain of the following functions.

a) $f(x) = \sin^{-1}(3x - 1)$

b) $j(x) = \frac{1}{x - \sqrt{x+2}}$

c) $f(x) = \ln|x + 3| - 5$

Solution

(a) $f(x) = \sin^{-1}(3x - 1)$

$D_f = \left[0, \frac{2}{3}\right]$

(b)

$j(x) = \frac{1}{x - \sqrt{x+2}}$

$x - \sqrt{x+2} \neq 0$

Also $x + 2 > 0$

So (via) $x - \sqrt{x+2} \neq 0$ we get $x = 2$

(c)

$(-\infty, -3) \cup (-3, \infty)$

All real numbers except -3.

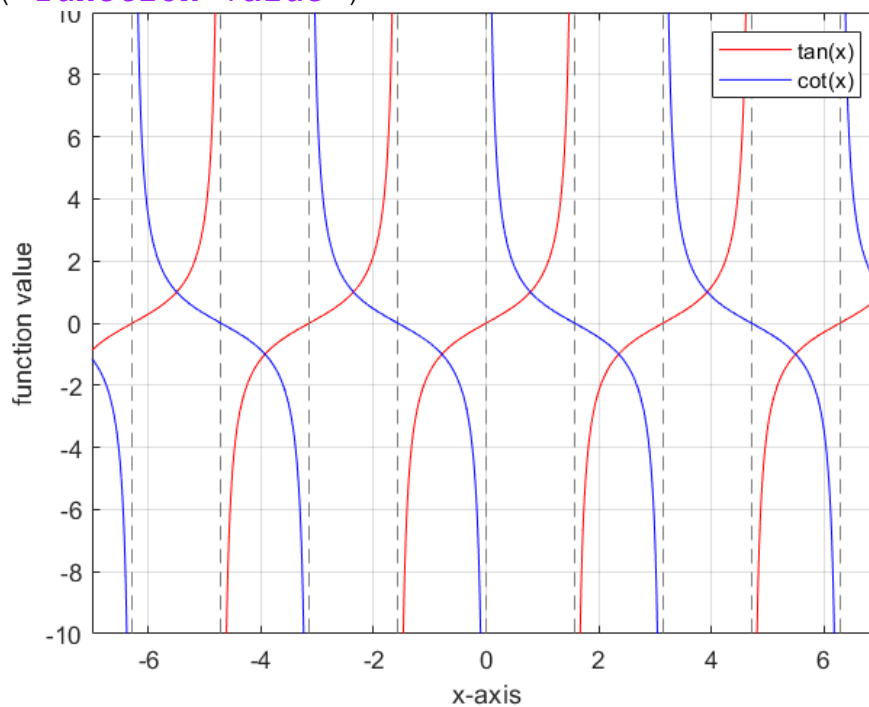
Question 15



Graph $y = \tan x$ and $y = \cot x$ together for $-7 \leq x \leq 7$. Comment on the behavior of $\cot x$ in relation to the signs and values of $\tan x$.

Solution

```
clear
clc
% question 15
syms x
y=tan(x);
z=cot(x);
figure(1)
fplot(y, 'r', [-7 7]);
hold on
fplot(z, 'b');
grid on
legend
xlabel('x-axis')
ylabel('function value')
```



When $\tan x$ is +ve $\cot x$ is also +ve and vice versa. When $\tan x$ is increasing $\cot x$ is decreasing.

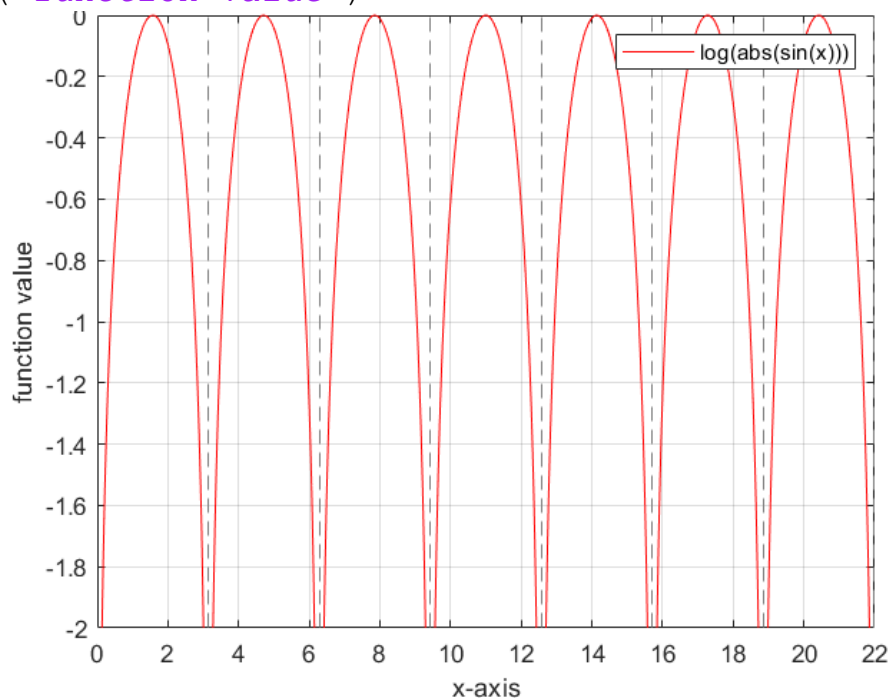
Question 16



Graph $y = \ln |\sin x|$ in the window $0 \leq x \leq 22$, $-2 \leq y \leq 0$.
Explain what you see. How could you change the formula to turn the arches upside down?

Solution

```
clear
clc
% question 16
syms x
y=log(abs(sin(x)));
figure(2)
fplot(y,'r',[0 22]);
ylim([-2 0]);
grid on
legend
xlabel('x-axis')
ylabel('function value')
```





National University

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Islamabad



One sees many curved lines that resemble arches. This is because this is a periodic function that repeats at every integer multiple of π . When $\sin x$ is close to zero, the function approaches negative infinity, then reaches a maximum of 0 since sine's max is 1. It repeats every π instead of 2π since negatives don't matter.

To display upside down arches just make the function negative (which will flip the function over the x -axis).